ASSIGNMENT No: 06 Title: Write a program for pre-processing of a text document such as stop word removal, stemming. Problem Statement: Write a program for pre-processing of a text document such as stop word removal, stemming.

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In [1]:
```
import nltk

nltk.download('punkt')
nltk.download('punkt_tab')
nltk.download('stopwords')

from nltk.corpus import stopwords
from nltk.tokenize import word_tokenize
from nltk.stem import PorterStemmer
```

```
[nltk_data] Downloading package punkt to
[nltk_data]     C:\Users\adity\AppData\Roaming\nltk_data...
[nltk_data]   Package punkt is already up-to-date!
[nltk_data] Downloading package punkt_tab to
[nltk_data]     C:\Users\adity\AppData\Roaming\nltk_data...
[nltk_data]   Package punkt_tab is already up-to-date!
[nltk_data] Downloading package stopwords to
[nltk_data]     C:\Users\adity\AppData\Roaming\nltk_data...
[nltk_data]   Package stopwords is already up-to-date!
```

In [2]:
```
text = """
Information retrieval is an important field of computer science.
It helps users find useful information from large collections of documents.
The system processes the documents and retrieves relevant information.
"""

print("Original Text:")
print(text)
```

```
Original Text:

Information retrieval is an important field of computer science.
It helps users find useful information from large collections of documents.
The system processes the documents and retrieves relevant information.
```

In [3]:
```
tokens = word_tokenize(text)

print("Tokens:")
print(tokens)
```

```
Tokens:
['Information', 'retrieval', 'is', 'an', 'important', 'field', 'of', 'computer', 'sc
ience', '.', 'It', 'helps', 'users', 'find', 'useful', 'information', 'from', 'larg
e', 'collections', 'of', 'documents', '.', 'The', 'system', 'processes', 'the', 'doc
uments', 'and', 'retrieves', 'relevant', 'information', '.']
```

In [5]:
```
stop_words = set(stopwords.words('english'))

filtered_words = []

for word in tokens:
    if word.lower() not in stop_words:
        filtered_words.append(word)

print("After Stop-Word Removal:")
print(filtered_words)
```

```
After Stop-Word Removal:
['Information', 'retrieval', 'important', 'field', 'computer', 'science', '.', 'help
s', 'users', 'find', 'useful', 'information', 'large', 'collections', 'documents',
'.', 'system', 'processes', 'documents', 'retrieves', 'relevant', 'information',
'.']
```

In [7]:
```
ps = PorterStemmer()

stemmed_words = []

for word in filtered_words:
    if word.isalpha():
        stemmed_words.append(ps.stem(word))
    else:
        stemmed_words.append(word)

print("After Stemming:")
print(stemmed_words)
print("Aditya Rade   BEAD23167")
```

```
After Stemming:
['inform', 'retriev', 'import', 'field', 'comput', 'scienc', '.', 'help', 'user', 'f
ind', 'use', 'inform', 'larg', 'collect', 'document', '.', 'system', 'process', 'doc
ument', 'retriev', 'relev', 'inform', '.']
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```

In [ ]: